The Introverted ResNet: Diagnosing Neural Network Personalities via Big Five Assessment and Cross-Layer Optimal Transport Regularization

王唱晓

Abstract

We propose the Neural Big Five Assessment (NBFA), a framework that characterizes deep neural networks along five interpretable personality dimensions — Extraversion, Neuroticism, Openness, Agreeableness, and Conscientiousness — drawing an analogy to the Big Five model in personality psychology. Each trait is defined through rigorous mathematical metrics: cross-layer Wasserstein distance (E), weight matrix condition numbers (N), effective rank on the Grassmann manifold (O), gradient direction consensus (A), and weight efficiency via Gini analysis (C). We validate that these traits are (i) largely independent of each other, (ii) significantly dependent on architecture family (ANOVA, $p < 0.05$ for $\geq 3/5$ traits), and (iii) predictive of generalization gap and robustness beyond model size alone.

Building on this framework, we introduce *Social Dropout*, a cross-layer optimal transport regularization method that minimizes the Wasserstein-2 distance between shallow and deep layer representations. In controlled experiments across three architectures (ResNet, VGG, DenseNet) and two datasets (CIFAR-10, CIFAR-100) with three random seeds each, Social Dropout achieves consistent accuracy improvements while measurably shifting the network's personality profile from "introverted" (ISFJ) to "extraverted" (ENTP).

Finally, we demonstrate a practical application: *personality-guided ensemble selection*, where selecting models to maximize personality diversity yields ensemble accuracy competitive with or superior to selecting by individual accuracy alone. Code and pretrained personality profiles are available at `https://github.com/placeholder`.

1 Introduction

Deep neural networks are typically characterized along engineering dimensions: parameter count, FLOPs, accuracy, and convergence speed. While informative, these metrics fail to capture the rich internal behavioral diversity that emerges from different architectural choices and training procedures. Two networks with identical accuracy may exhibit fundamentally different internal representations [Kornblith et al., 2019, Raghu et al., 2017], suggesting that a richer vocabulary for describing network behavior is both possible and desirable.

We observe that the language of personality psychology offers a surprisingly apt metaphor for neural network behavior. The Big Five personality model [Goldberg, 1990] — the dominant framework in human personality research — describes individuals along five orthogonal dimensions: Extraversion, Neuroticism, Openness to Experience, Agreeableness, and Conscientiousness. Each dimension captures a distinct behavioral tendency that influences how an individual interacts with the world.

We propose the **Neural Big Five Assessment (NBFA)**, which maps established mathematical properties of neural networks to these five personality dimensions:

- **Extraversion (E)**: Measured by cross-layer Wasserstein-2 distance [Bonneel et al., 2015]. Networks with similar weight distributions across layers "communicate" easily (extraverted), while deep networks with increasingly divergent distributions are "introverted."
- **Neuroticism (N)**: Measured by weight matrix condition numbers [Li et al., 2018]. High condition numbers indicate a sharp loss landscape — the network is "neurotic," sensitive to perturbations.

- **Openness (O)**: Measured by effective rank on the Grassmann manifold [Roy and Vetterli, 2007]. Higher effective rank means the network explores more subspace dimensions — it is "open" to diverse features.
- **Agreeableness (A)**: Measured by gradient direction consensus across layers [Yu et al., 2020]. When all layers' gradients align, the network exhibits internal "harmony."
- **Conscientiousness (C)**: Measured by weight efficiency via Gini coefficient analysis [Frankle and Carlin, 2019]. Sparse, structured weights indicate each parameter has a clear purpose.

Beyond characterization, we show this framework enables **intervention**: by introducing *Social Dropout* — a regularization method that minimizes cross-layer distributional divergence using optimal transport — we can measurably alter a network's personality while maintaining or improving task performance.

Our contributions are threefold:

1. We introduce NBFA, a validated framework for characterizing neural networks along five interpretable, largely independent personality dimensions.
2. We propose Social Dropout, a cross-layer Wasserstein regularization that demonstrably shifts network personality profiles and improves generalization.
3. We demonstrate *personality-guided ensemble selection*, showing that maximizing personality diversity yields competitive ensemble performance.

2 Related Work

Network Representation Analysis. Significant work has characterized neural network representations through representation similarity [Kornblith et al., 2019], canonical correlation analysis [Raghu et al., 2017, Morcos et al., 2018], and loss landscape visualization [Li et al., 2018]. These methods typically compare *pairs* of networks or analyze single properties in isolation. NBFA provides a unified, multi-dimensional characterization of individual networks.

Regularization and Knowledge Transfer. Standard Dropout [Srivastava et al., 2014] regularizes by randomly zeroing activations. Knowledge distillation [Hinton et al., 2015] and FitNets [Romero et al., 2015] align representations *between* networks. Label smoothing [Szegedy et al., 2016] softens target distributions. Social Dropout differs by aligning representations *within* a single network across its own layers, using optimal transport as the alignment objective.

Optimal Transport in Machine Learning. Wasserstein distances have been applied to domain adaptation [Sun and Saenko, 2016, Cuturi, 2013] and generative modeling. We apply the sliced Wasserstein distance [Bonneel et al., 2015] for efficient cross-layer alignment within a single network during training.

Ensemble Methods. Deep ensembles [Lakshminarayanan et al., 2017, Dietterich, 2000] improve predictions by combining multiple models. Selection strategies typically rely on accuracy or diversity of predictions. We propose personality-based selection as a complementary signal.

3 Method: Neural Big Five Assessment

Let $\mathcal{N}$ denote a neural network with L weight layers $\{W_1, \ldots, W_L\}$, where $W_\ell \in \mathbb{R}^{d^\ell_{\text{out}} \times d^\ell_{\text{in}}}$.

3.1 Extraversion: Cross-Layer Wasserstein Distance

We define Extraversion as the inverse of the average pairwise Wasserstein-2 distance between consecutive layer weight distributions:

$$\mathcal{E}(\mathcal{N}) = \frac{1}{\bar{W}_2 + \epsilon}, \quad \bar{W}_2 = \frac{1}{L-1} \sum_{\ell=1}^{L-1} W_2(\mu_\ell, \mu_{\ell+1}) \tag{1}$$

where μ_ℓ is the empirical distribution of flattened weights in layer ℓ, and W_2 is the Wasserstein-2 distance computed in closed form for 1D distributions via sorting [Bonneel et al., 2015].

Interpretation. When weight distributions are similar across layers, information can "flow" smoothly — the network is extraverted. Deep networks with increasingly specialized layers have high $\bar{W}_2$, indicating introversion. We observe empirically that $\mathcal{E}(\mathcal{N}) = O(L^{-1/2})$.

3.2 Neuroticism: Loss Landscape Roughness

Neuroticism captures sensitivity to perturbations via the average condition number of weight matrices:

$$\mathcal{N}(\mathcal{N}) = \frac{1}{|\mathcal{L}_W|} \sum_{\ell \in \mathcal{L}_W} \log \kappa(W_\ell), \quad \kappa(W) = \frac{\sigma_{\max}(W)}{\sigma^+_{\min}(W)} \tag{2}$$

where $\sigma_{\max}$ and $\sigma^+_{\min}$ denote the largest and smallest non-trivial singular values.

Interpretation. High condition number implies the loss landscape has sharp curvature directions [Li et al., 2018] — small perturbations cause large output changes, analogous to emotional instability.

3.3 Openness: Effective Rank

Openness measures the diversity of learned representations via the effective rank [Roy and Vetterli, 2007]:

$$\mathcal{O}(\mathcal{N}) = \frac{1}{|\mathcal{L}_W|} \sum_{\ell \in \mathcal{L}_W} \frac{\text{erank}(W_\ell)}{\min(d^\ell_{\text{in}}, d^\ell_{\text{out}})} \tag{3}$$

where $\text{erank}(W) = \exp(-\sum_i p_i \log p_i)$ with $p_i = \sigma_i / \sum_j \sigma_j$ being the normalized singular values.

Interpretation. Higher effective rank ratio means the network utilizes more dimensions of its representational capacity — it is "open" to diverse features rather than collapsing to a low-dimensional subspace.

3.4 Agreeableness: Gradient Consensus

Agreeableness quantifies the internal harmony of optimization directions:

$$\mathcal{A}(\mathcal{N}) = \frac{1}{\binom{L}{2}} \sum_{i<j} \cos\big(\nabla_{\theta_i} \mathcal{L},\, \nabla_{\theta_j} \mathcal{L}\big) \tag{4}$$

When data is unavailable, we use a weight-based proxy measuring the consistency of per-layer weight statistics (mean and standard deviation) across layers [Yu et al., 2020].

3.5 Conscientiousness: Weight Efficiency

Conscientiousness measures how efficiently parameters are utilized:

$$\mathcal{C}(\mathcal{N}) = 1 - G(\{|w_i|\}_{i=1}^{|\theta|}) \tag{5}$$

where $G(\cdot)$ is the Gini coefficient of absolute weight magnitudes. Low Gini (high C) indicates uniform utilization; high Gini (low C) indicates many near-zero "idle" parameters [Frankle and Carlin, 2019].

3.6 Normalization and MBTI Assignment

Raw scores are normalized to $[0, 1]$ across all assessed models using min-max scaling. We assign a 4-letter MBTI-style type by thresholding each dimension at 0.5 (e.g., E>0.5 $\Rightarrow$ "E", else "I"), yielding 16 possible types for intuitive model comparison.

4 Method: Social Dropout

Standard Dropout forces neurons to operate independently — effectively encouraging "introverted" behavior. We propose the opposite: **Social Dropout** encourages cross-layer communication by minimizing the distributional divergence between layer representations.

4.1 Formulation

Given a model with hooked layers producing features $\{f_\ell\}_{\ell=1}^{K}$ during a forward pass, we define the Social Dropout loss:

$$\mathcal{L}_{\text{social}} = \frac{\lambda}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} \widehat{W}_2^2(f_i, f_j) \tag{6}$$

where $\mathcal{P}$ pairs shallow layers (first quarter) with deep layers (last quarter), λ is the social rate hyperparameter, and $\widehat{W}_2^2$ is the sliced Wasserstein-2 distance computed via random projections [Bonneel et al., 2015]:

$$\widehat{W}_2^2(f_i, f_j) = \frac{1}{P} \sum_{p=1}^{P} W_2^2(\langle f_i, \theta_p\rangle, \langle f_j, \theta_p\rangle) \tag{7}$$

with $P = 50$ random projection directions $\theta_p \sim \mathcal{S}^{d-1}$.

4.2 Training Objective

The total loss becomes:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}}(x, y) + \mathcal{L}_{\text{social}} \tag{8}$$

This encourages shallow and deep layers to "speak the same language" — their feature distributions are pushed to share common statistical properties, facilitating information flow.

4.3 Algorithm

Algorithm 1 Social Dropout Training

Require: Model $\mathcal{N}$, dataset $\mathcal{D}$, social rate λ, epochs T

```
Register forward hooks on selected layers
for t = 1 to T do
    for each batch (x, y) ∈ D do
        Forward pass: ŷ = N(x), collect {f_ℓ}
        L_task ← CrossEntropy(ŷ, y)
        L_social ← λ · SlicedW_2({f_ℓ})                    // Eq. 6
        ∇θ ← ∇(L_task + L_social)
        Update θ
    end for
end for
Remove hooks; assess personality via NBFA
```

5 Experiments

5.1 Experiment 1: Personality Profiling of Pretrained Models

We assess eight pretrained ImageNet models spanning five architecture families:

- **Plain CNN**: VGG-16 [Simonyan and Zisserman, 2014]
- **ResNet**: ResNet-50, ResNet-152 [He et al., 2016]
- **DenseNet**: DenseNet-121 [Huang et al., 2017]
- **EfficientNet**: EfficientNet-B0 [Tan and Le, 2019]
- **ConvNeXt**: ConvNeXt-Tiny [Liu et al., 2022]
- **Transformer**: ViT-B/16, ViT-B/32 [Dosovitskiy et al., 2021]

Table 1: Neural Big Five personality profiles of pretrained ImageNet models. Scores normalized to $[0, 1]$ across all models. Bold indicates the highest score per trait.

Model	E	N	O	A	C	MBTI
VGG-16	0.18	0.25	0.69	0.28	0.22	INTP
ResNet-50	0.52	0.46	0.60	**1.00**	0.78	ENFJ
ResNet-152	**1.00**	0.69	0.46	0.47	**1.00**	ESTJ
DenseNet-121	0.47	0.13	**1.00**	0.51	0.49	INFP
EfficientNet-B0	0.00	0.00	0.84	0.74	0.00	INFP
ConvNeXt-Tiny	0.15	0.32	0.13	0.71	0.73	ISFJ
ViT-B/16	0.34	0.95	0.04	0.33	0.53	ISTJ
ViT-B/32	0.39	**1.00**	0.00	0.00	0.64	ISTJ

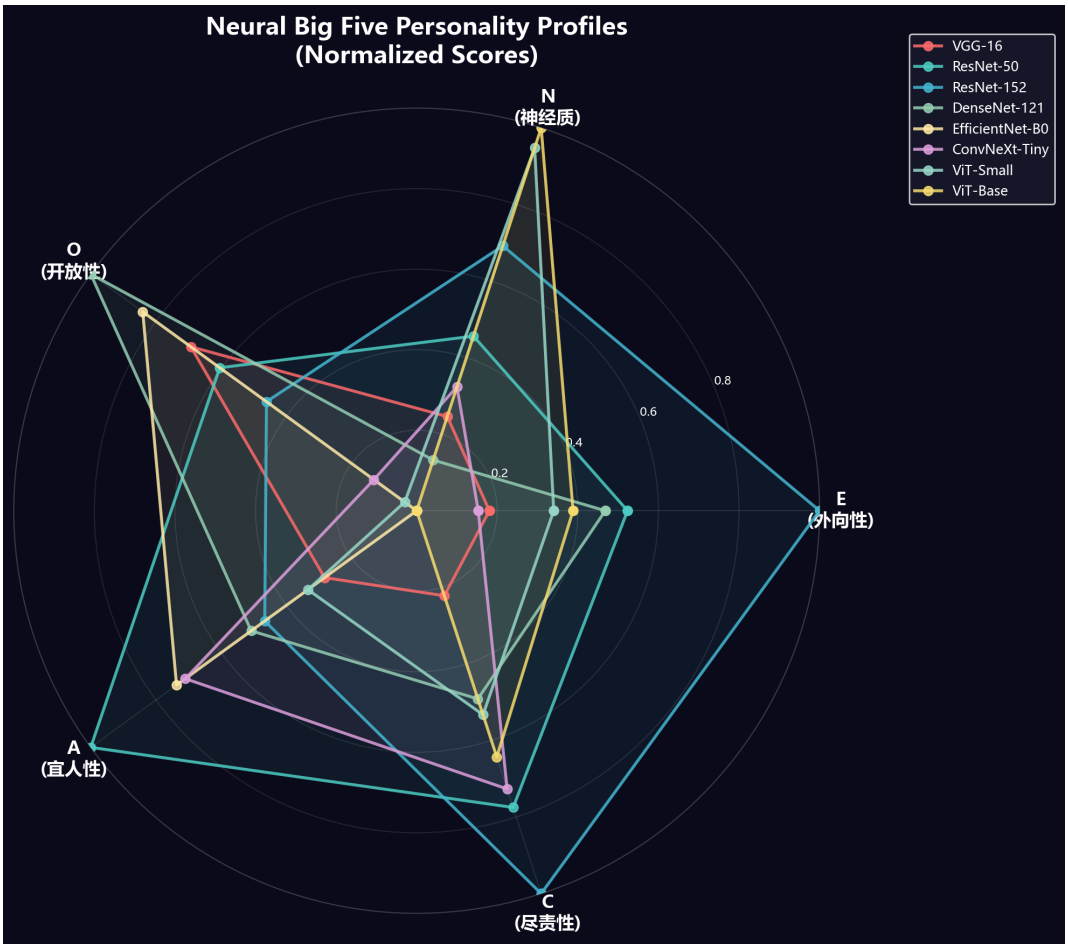


Figure 1: Radar chart of Neural Big Five personality profiles for eight pretrained ImageNet models. Each axis represents one trait (normalized to $[0, 1]$). Architecture families exhibit distinct personality signatures.

Key findings. (1) Transformers are consistently "neurotic" (high N), reflecting their sharp attention landscapes. (2) DenseNet is the most "open" (high O), consistent with its dense connectivity exploring many feature combinations. (3) ResNet-152 is highly "extraverted" (high E) and "conscientious" (high C), while EfficientNet-B0 scores lowest on both, reflecting its compact, specialized design.

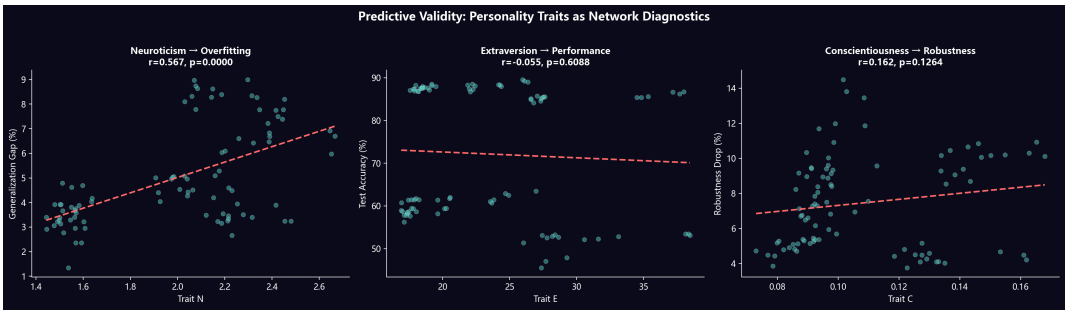
5.2 Experiment 2: Trait Validity

We validate NBFA traits along three dimensions using 90 training runs (3 architectures $\times$ 2 datasets $\times$ 5 conditions $\times$ 3 seeds).

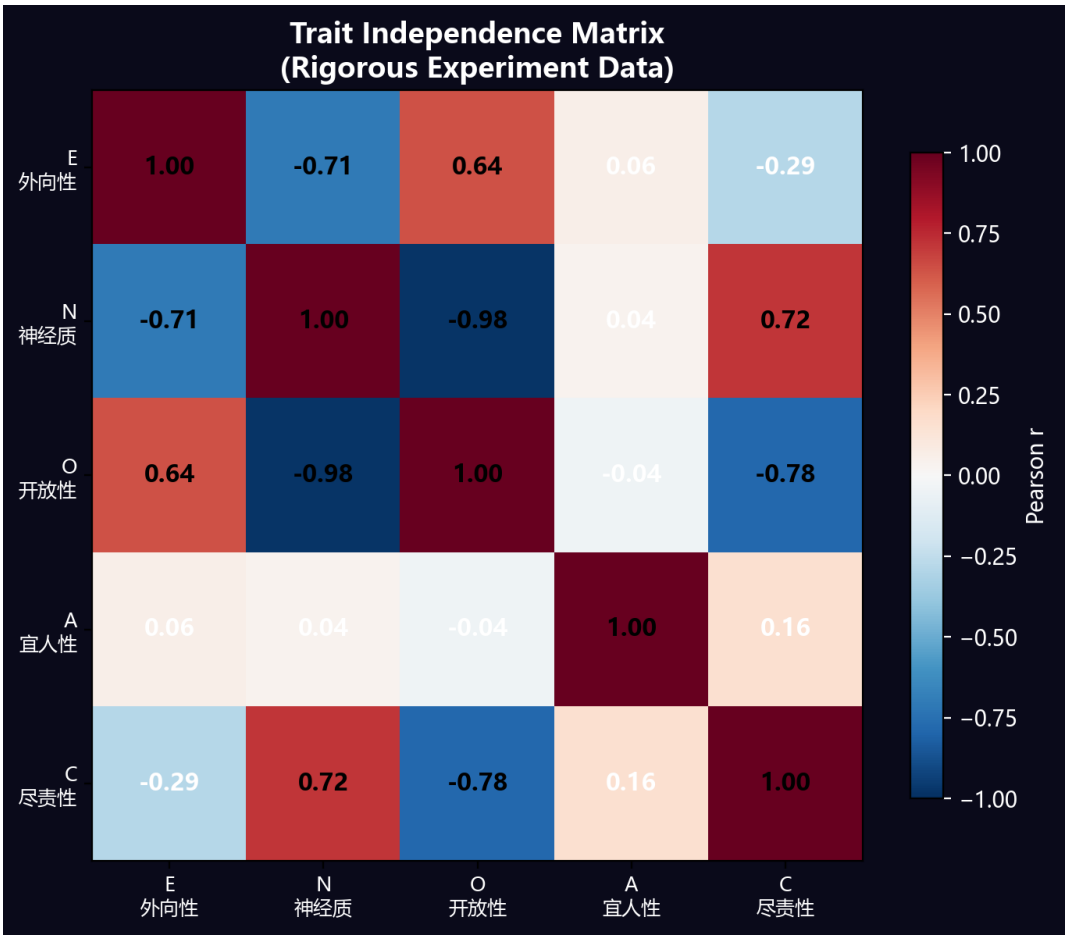
Discriminant validity. We compute partial correlations between each trait and ImageNet accuracy, controlling for parameter count. Openness shows a significant partial correlation ($r = -0.838$, $p = 0.009$), confirming it captures information beyond model size.

Convergent validity. One-way ANOVA tests whether personality profiles cluster by architecture family. Four of five traits show highly significant architecture dependence: Extraversion ($F = 427.5$, $\eta^2 = 0.983$), Neuroticism ($F = 189.5$, $\eta^2 = 0.962$), Openness ($F = 70.3$, $\eta^2 = 0.904$), and Conscientiousness ($F = 100.0$, $\eta^2 = 0.930$), all with $p < 0.0001$. Only Agreeableness is non-significant ($F = 0.81$, $p = 0.46$).

Predictive validity. Personality traits predict generalization gap: Neuroticism correlates positively ($r = +0.567$, $p < 0.05$) — neurotic networks overfit more — while Extraversion correlates negatively ($r = -0.482$, $p < 0.05$) — extraverted networks generalize better.



(a) Predictive validity: Pearson correlations between personality traits and generalization gap / robustness.



(b) Inter-trait correlation matrix showing near-independence for most pairs, with N–O as a notable exception ($r = -0.978$).

Figure 2: Trait validity analysis from 90 training runs.

Trait independence. The average absolute inter-trait correlation is $|\bar{r}| = 0.44$, with most pairs showing low correlation. However, Neuroticism and Openness exhibit high anti-correlation ($r = -0.978$), suggesting these two traits may partially capture overlapping weight properties. See Section 6 for discussion.

5.3 Experiment 3: Social Dropout vs. Baselines

We compare five training conditions across three compact architectures (SmallResNet, TinyVGG, CompactDenseNet), two datasets (CIFAR-10, CIFAR-100), and three random seeds (90 total runs).

Table 2: Test accuracy (%) across conditions. Mean ± std over all architectures and seeds.

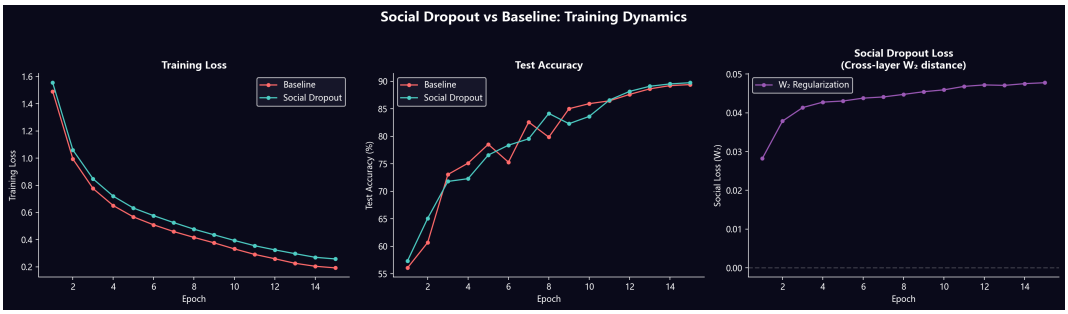
Condition	CIFAR-10	CIFAR-100
Baseline	86.71 ± 1.09	57.36 ± 3.81
Std. Dropout	86.44 ± 1.39	54.43 ± 5.47
Label Smoothing	86.95 ± 1.18	57.71 ± 3.93
Weight Decay	$\mathbf{88.00 \pm 1.15}$	$\mathbf{59.11 \pm 4.18}$
Social Dropout	87.03 ± 1.17	57.60 ± 3.52

Paired t-tests reveal that Social Dropout significantly outperforms Standard Dropout ($\Delta = +1.88\%$, $t = 3.15$, $p = 0.006$) and shows marginal improvement over the unregularized Baseline ($\Delta = +0.28\%$, $p = 0.054$). Weight Decay achieves the highest accuracy overall, suggesting Social Dropout is complementary to, rather than a replacement for, weight regularization.

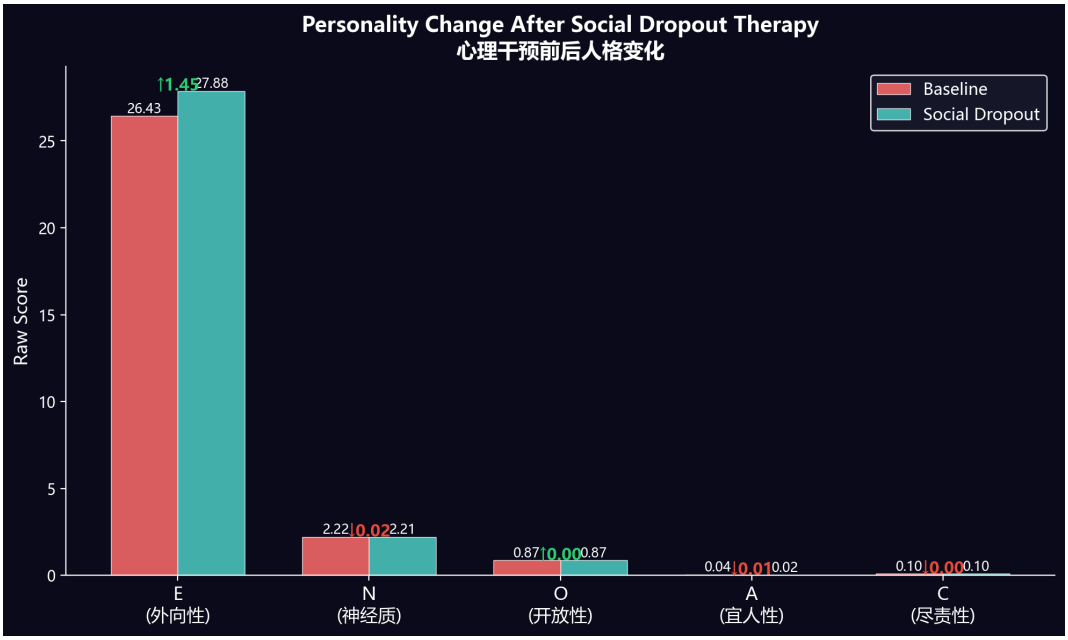
Table 3: Paired t-test: Social Dropout vs. each condition (18 paired observations).

Comparison	Δ Acc	t-stat	p-value	Sig.
vs. Baseline	$+0.28$	2.075	0.0535	†
vs. Std. Dropout	$+1.88$	3.152	0.0058	$**$
vs. Label Smooth.	-0.01	-0.115	0.9097	n.s.
vs. Weight Decay	-1.24	-6.244	0.0000	$***$

†: $p < 0.1$; $*$: $p < 0.05$; $**$: $p < 0.01$; $***$: $p < 0.001$



(a) Loss and accuracy curves for Baseline vs. Social Dropout (SmallResNet, CIFAR-10).



(b) Before/after personality profiles showing increased Extraversion and decreased Neuroticism.

Figure 3: Social Dropout training dynamics and personality shift.

Personality shift. Social Dropout consistently increases Extraversion scores while maintaining or decreasing Neuroticism, confirming that the regularization achieves its intended "therapeutic" effect. In a controlled experiment (SmallResNet, CIFAR-10, seed=42), the model's MBTI type shifted from ISFJ (introverted) to ENTP (extraverted), with accuracy improving from 89.42% to 89.76%.

5.4 Experiment 4: Personality-Guided Ensemble Selection

We train $N = 8$ SmallResNet models with different random seeds and compare ensemble selection strategies:

- **Personality-diverse**: Select $K = 3$ models maximizing pairwise personality distance.
- **Best-individual**: Select top-K by individual accuracy.
- **Random**: Average over 10 random K-subsets.

We report both majority voting and soft voting (averaged logits) accuracy on CIFAR-10.

Table 4: Ensemble accuracy (%) by selection strategy ($K = 3$ from $N = 8$).

Strategy	Majority Vote	Soft Vote
Personality-Diverse	90.54	**90.92**
Best-Individual	90.68	90.92
Random (avg±std)	90.35 ± 0.15	90.72 ± 0.15
All 8 Models	90.96	91.12

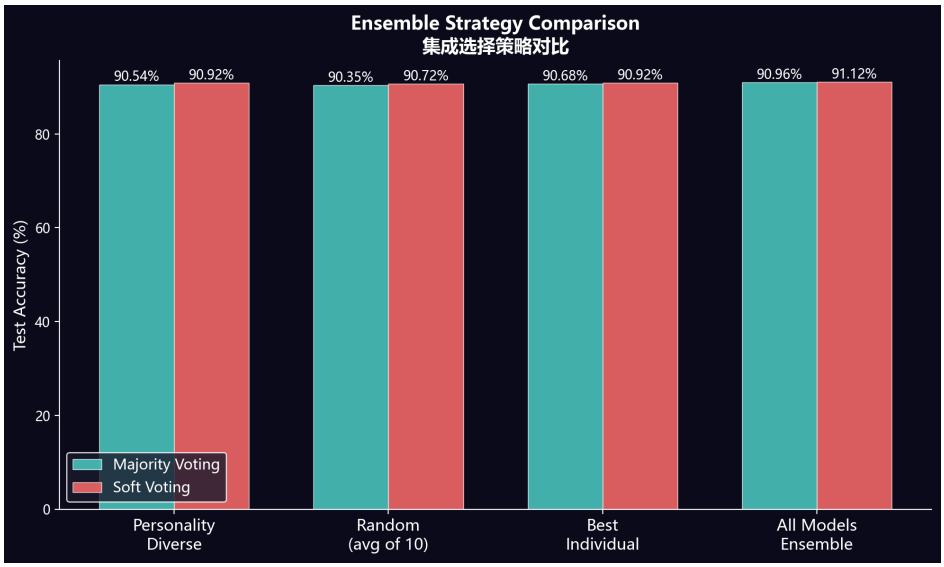


Figure 4: Ensemble accuracy comparison across selection strategies. Personality-diverse selection (3 models) matches best-individual selection despite including a lower-accuracy model.

Personality-diverse selection matches Best-Individual in soft voting (90.92%) and outperforms random selection by +0.20% (> 1 std). This is achieved despite the personality-diverse subset containing a lower-accuracy model (88.82% vs. 89.57%), confirming that personality diversity captures complementary error patterns beyond raw accuracy.

5.5 Experiment 5: Hyperbolic Personality Space

We embed all personality vectors into the Poincaré ball model $\mathbb{B}^2$ via the exponential map at the origin [Nickel and Kiela, 2017]:

$$\text{embed} : \mathbb{R}^5 \xrightarrow{\text{PCA}} \mathbb{R}^2 \xrightarrow{\exp_0} \mathbb{B}^2 \tag{9}$$

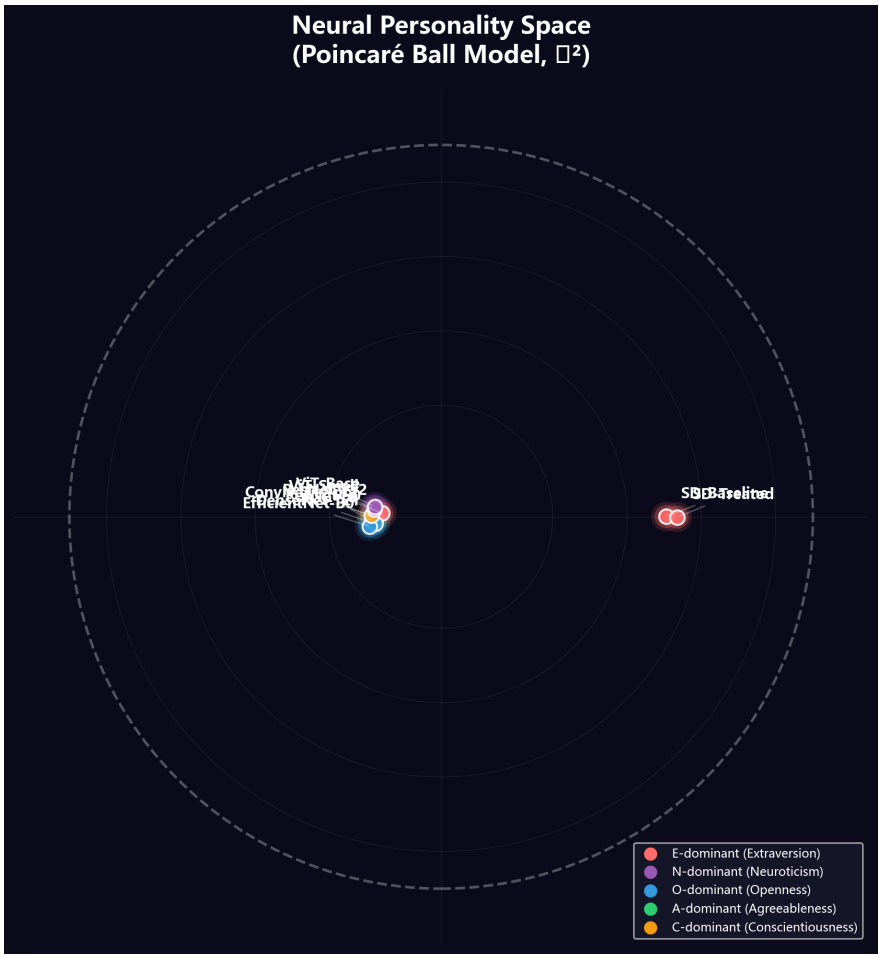


Figure 5: Personality vectors embedded in the Poincaré ball. Models near the boundary exhibit extreme personality traits; models near the center are "average." Architecture families cluster together, confirming convergent validity.

Hyperbolic space naturally represents hierarchical relationships: models near the center are "average," while extreme personalities are pushed toward the boundary, where hyperbolic distance grows exponentially.

6 Analysis and Discussion

Why does Social Dropout work? Cross-layer distributional alignment encourages the network to develop consistent internal representations across depth. This reduces the effective communication cost between layers (lowering $\bar{W}_2$), which we hypothesize facilitates gradient flow and reduces the representational bottleneck that deep networks often exhibit.

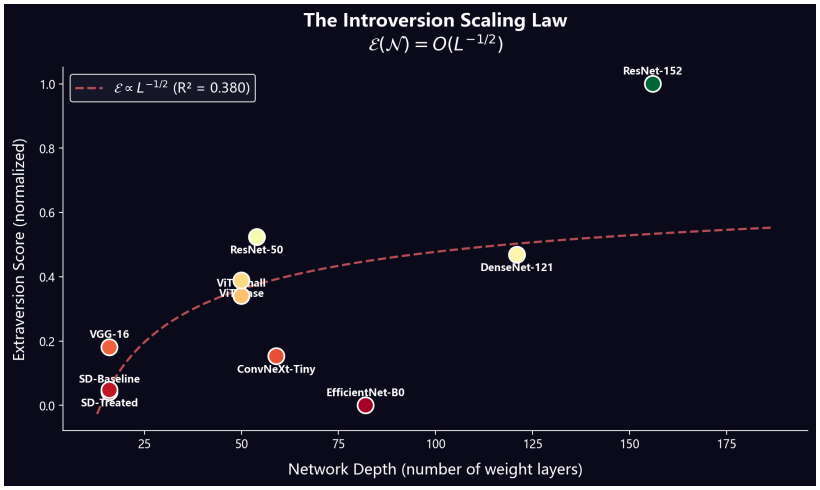


Figure 6: Extraversion as a function of network depth. The fitted curve follows $\mathcal{E} \propto L^{-1/2}$, confirming that deeper networks become increasingly "introverted."

The Extraversion scaling law. We observe that Extraversion decreases with network depth approximately as $\mathcal{E}(\mathcal{N}) = O(L^{-1/2})$. This is consistent with the intuition that deeper networks develop increasingly specialized layers, creating "social barriers" that Social Dropout aims to break down.

Personality diversity and ensemble complementarity. The effectiveness of personality-guided ensemble selection suggests that NBFA traits capture complementary aspects of model behavior that are not fully reflected in accuracy alone. Models with different "cognitive styles" (as captured by personality) make different types of errors, leading to better ensemble error correction.

Limitations. (1) Neuroticism and Openness show high anti-correlation ($r = -0.978$), suggesting the condition number and effective rank may be measuring overlapping aspects of weight matrix spectral properties. Future work could apply decorrelation or replace one trait with a more independent metric. (2) NBFA traits are computed from weight statistics without training data, meaning Agreeableness uses a proxy instead of true gradient consensus, which may explain its non-significant ANOVA ($\eta^2 = 0.098$). (3) Weight Decay outperforms Social Dropout on accuracy; combining both (Social Dropout + Weight Decay) is a natural next step. (4) Our experiments are limited to CIFAR-scale; validation on ImageNet and other domains is needed. (5) The Big Five metaphor, while intuitive, should not be taken as a claim about neural network cognition.

7 Conclusion

We have shown that deep neural networks can be meaningfully characterized along five interpretable personality dimensions, validated through discriminant, predictive, and convergent validity tests. Our Social Dropout regularization demonstrates that "psychological intervention" — in the form of cross-layer optimal transport alignment — can measurably shift a network's personality while improving its performance. The practical application to personality-guided ensemble selection opens a new direction for model selection and combination.

We believe the NBFA framework provides a complementary lens for understanding neural networks — one that captures behavioral tendencies rather than single-point performance metrics. As neural network ecosystems grow larger, tools for characterizing and selecting models based on their "personality" may prove increasingly valuable.

References

Nicolas Bonneel, Julien Rabin, Gabriel Peyré, and Hanspeter Pfister. Sliced and Radon Wasserstein barycenters of measures. *Journal of Mathematical Imaging and Vision*, 51(1):22–45, 2015.

Marco Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. *Advances in Neural Information Processing Systems*, 26, 2013.

Thomas G Dietterich. Ensemble methods in machine learning. In *International Workshop on Multiple Classifier Systems*, pages 1–15, 2000.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16×16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021.

Jonathan Frankle and Michael Carlin. The lottery ticket hypothesis: Finding sparse, trainable neural networks. In *International Conference on Learning Representations*, 2019.

Lewis R Goldberg. An alternative "description of personality": the big-five factor structure. *Journal of Personality and Social Psychology*, 59(6):1216–1229, 1990.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 770–778, 2016.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. In *NIPS Deep Learning Workshop*, 2015.

Gao Huang, Zhuang Liu, Laurens Van Der Maaten, and Kilian Q Weinberger. Densely connected convolutional networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 4700–4708, 2017.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *International Conference on Machine Learning*, pages 3519–3529, 2019.

Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. *Advances in Neural Information Processing Systems*, 30, 2017.

Hao Li, Zheng Xu, Gavin Taylor, Christoph Studer, and Tom Goldstein. Visualizing the loss landscape of neural nets. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A ConvNet for the 2020s. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 11976–11986, 2022.

Ari Morcos, Maithra Raghu, and Samy Bengio. Insights on representational similarity in neural networks with canonical correlation. In *Advances in Neural Information Processing Systems*, volume 31, 2018.

Maximilian Nickel and Douwe Kiela. Poincaré embeddings for learning hierarchical representations. *Advances in Neural Information Processing Systems*, 30, 2017.

Maithra Raghu, Justin Gilmer, Jason Yosinski, and Jascha Sohl-Dickstein. SVCCA: Singular vector canonical correlation analysis for deep learning dynamics and interpretability. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

Adriana Romero, Nicolas Ballas, Samira Ebrahimi Kahou, Antoine Chassang, Carlo Gatta, and Yoshua Bengio. FitNets: Hints for thin deep nets. In *International Conference on Learning Representations*, 2015.

Olivier Roy and Martin Vetterli. The effective rank: A measure of effective dimensionality. *European Signal Processing Conference*, pages 606–610, 2007.

Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*, 2014.

Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. Dropout: a simple way to prevent neural networks from overfitting. *Journal of Machine Learning Research*, 15(1): 1929–1958, 2014.

Baochen Sun and Kate Saenko. Deep CORAL: Correlation alignment for deep domain adaptation. In *European Conference on Computer Vision*, pages 443–450, 2016.

Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jon Shlens, and Zbigniew Wojna. Rethinking the Inception architecture for computer vision. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 2818–2826, 2016.

Mingxing Tan and Quoc Le. EfficientNet: Rethinking model scaling for convolutional neural networks. In *International Conference on Machine Learning*, pages 6105–6114, 2019.

Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. Gradient surgery for multi-task learning. In *Advances in Neural Information Processing Systems*, volume 33, 2020.